GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A drawing has a 35° angle. A scale copy is made at k = 3. What is the matching angle in the copy? *1 pt*

- ○ A. 105°
- ○ B. 35°
- ○ C. 11.67°
- ○ D. 38°

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. Find k, and explain why k is not 9. *1 pt*

6. A floor plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square metres. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimetres is 9 in? *1 pt*

10. 1 cm = 0.39 in. How many inches is 70 cm? *1 pt*

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 4 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

13. 1 gal = 3.785 L. How many litres is 5 gal? *1 pt*

14. 1 yd = 3 ft. A rug is 4 yd^2 in area. How many square feet is that? *1 pt*

- ○ A. 12 ft^2
- ○ B. 9 ft^2
- ○ C. 36 ft^2
- ○ D. 24 ft^2

15. Rate on 5 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

b. How many US dollars is 23 euros? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car does 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

b. It starts a 1,000-mi trip with a full tank. How many tanks of gas does the trip take? *1 pt*

17. A recipe for 4 servings uses 1½ cups of rice.

a. How much rice for 10 servings? *1 pt*

b. With 6 cups on hand, how many servings can be made? *1 pt*

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A 200-km drive. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how long? Explain. *1 pt*